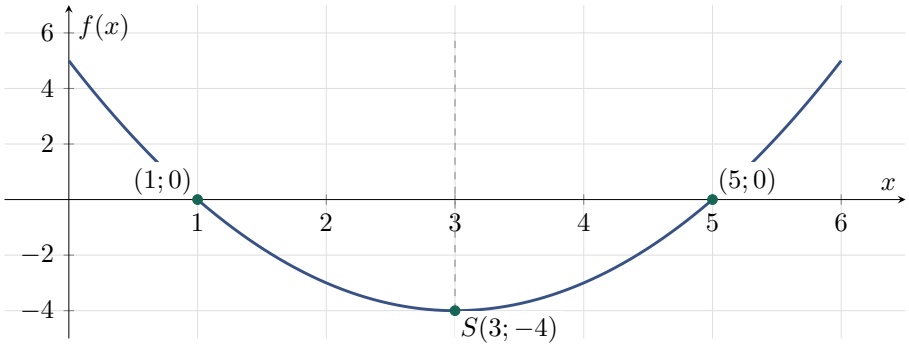
2.1 Three forms, three readings

Definition and results

A quadratic function is written $f(x) = ax^2 + bx + c$ with $a \neq 0$. The expanded form makes it easy to calculate $f(0)$; the vertex form $a(x-\alpha)^2 + \beta$ reveals the vertex and the extremum; a factorised form $a(x-r_1)(x-r_2)$ reveals the roots and the sign. The roots are the solutions of $f(x) = 0$.

Status and justification. Completing the square relies on $x^2 + 2ux = (x+u)^2 - u^2$. The existence of the vertex form is proved by expansion; its general computation is not an expected routine.

Worked example. $f(x) = x^2 - 6x + 5 = (x-3)^2 - 4 = (x-1)(x-5)$. The minimum is -4 at 3; the roots are 1 and 5; the y-intercept is 5.



Graph of $f(x) = (x-3)^2 - 4$. The two roots and the vertex are marked; the axis of symmetry is the line $x = 3$.

Watch out

A factorised expression is not obtained by putting each term separately in brackets.

Immediate application. Give three forms of $g(x) = x^2 + 2x - 3$. Read off its roots and its minimum.

2.2 Discriminant and solving

Definition and results

For $ax^2+bx+c=0$ with $a\neq 0$, let $\Delta=b^2-4ac$. If $\Delta<0$, there is no real root; if $\Delta=0$, there is one root $-\dfrac{b}{2a}$; if $\Delta>0$, there are two roots $\dfrac{-b-\sqrt{\Delta}}{2a}$ and $\dfrac{-b+\sqrt{\Delta}}{2a}$. To order the roots, take the sign of a into account.

Status and justification. Multiplying the equation by $4a\neq 0$ gives $(2ax+b)^2=\Delta$. A square is positive or zero: the three cases follow. If $\Delta>0$, $2ax+b=\pm\sqrt{\Delta}$. Splitting into cases avoids taking the square root of a negative number.

Worked example. $2x^2+x-3=0$: $\Delta=1+24=25$; $x=\dfrac{-1\pm 5}{4}$, that is 1 and $-\dfrac{3}{2}$. Check: $(x-1)(2x+3)=2x^2+x-3$.

Watch out

The denominator is $2a$, not 2; the sign of b is kept with its brackets in b^2.

Immediate application. Solve $3x^2-6x+3=0$ and $x^2+2x+5=0$.

2.3 Factorising without automating the discriminant

Definition and results

If two real roots exist, their sum is $-\dfrac{b}{a}$ and their product is $\dfrac{c}{a}$. An obvious root or a special product often allows a shorter factorisation. The polynomials of degree 2 that vanish at two distinct reals r and s are exactly $a(x-r)(x-s)$ with $a\neq 0$.

Status and justification. Expanding $a(x-r)(x-s)=ax^2-a(r+s)x+ars$, identifying the coefficients gives the relations. Conversely, if r and s have this sum and this product, the expansion recovers the polynomial.

Worked example. $x^2-7x+12$: 3 and 4 have sum 7 and product 12. Hence $(x-3)(x-4)$. To impose in addition $f(0)=24$, we need $12a=24$, so $a=2$.

Watch out

The sum of the roots is $-\dfrac{b}{a}$; do not forget the minus sign. A single root is not enough to determine a.

Immediate application. Determine the polynomial f of degree 2 such that $f(1)=f(4)=0$ and $f(0)=-8$.

2.4 Sign, inequalities and optimisation

Definition and results

For two distinct roots $r < s$, the quadratic has the sign of a outside $[r; s]$ and the opposite sign between the roots. It is zero at the roots. With no real root, it keeps the sign of a. With a double root, it keeps the sign of a away from that root. Optimisation is done on the domain of the problem, which may limit access to the vertex.

Status and justification. The sign table of the factors of $a(x - r)(x - s)$ gives the result. The vertex form compares $f(x)$ with β through the sign of $a(x - \alpha)^2$.

Worked example. $-2(x - 1)(x - 5) \geq 0$ is equivalent to $x \in [1; 5]$. For $A(x) = x(12 - x)$ on $[0; 12]$, $A(x) = 36 - (x - 6)^2 \leq 36$, with equality at 6. A rectangle with perimeter 24 m therefore has a maximum area of 36 m^2 when it is a square.

Watch out

In a strict inequality, the roots are excluded. Check the geometric bounds before concluding.

Immediate application. Solve $x^2 - 4x - 5 < 0$, then maximise $x(8 - x)$ on $[0; 3]$.